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Pioneering Production Rejuvenation: Coiled Tubing Hybrid Fiber Optics Redefines Well Intervention Efficiency and Restores Production After 8 Years of Inactivity in Shut-In Wells — A Landmark Recovery in Kuwait's Mature Fields in the North

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Abstract

This paper presents a transformative workflow for economically reviving production from mature oil fields, demonstrated through the first successful deployment of Coiled Tubing Hybrid Fiber Optics Technology (CT HFOT) in North Kuwait. Addressing the persistent challenge of restoring productivity in long-term suspended wells, the intervention set a new benchmark for operational efficiency, safety, and production rejuvenation in the region's mature assets.

North Kuwait has faced an increasing inventory of suspended wells due to early water breakthrough, mechanical issues, and high intervention costs. Traditional workover and stimulation methods proved insufficient in economically reactivating these wells. To overcome these limitations, the reservoir management team launched a comprehensive campaign focused on reactivating inactive wells through a workflow combining advanced diagnostics, optimized execution, and precise fluid placement.

The intervention strategy integrated conventional coiled tubing (CT) methods with fiber-optic-enabled workflows, redefining rig-based intervention protocols. A key innovation was the application of CT HFOT, eliminating the need for multiple CT reels and downhole batteries, typical challenges in extended well interventions. By embedding optical fibers and insulated conductor cables in the CT string, real-time downhole data acquisition was enabled, providing continuous feedback on gamma ray, pressure, temperature, and axial forces. This real-time insight optimized perforation accuracy, cleanout efficiency, and fluid placement while reducing rig-up time and simplifying operations.

A landmark case targeted a horizontal well drilled in 2015, originally completed as a Mauddud producer with a 4-½ in. inflow control device (ICD) lower completion. Dormant for over eight years, this well became the first CT HFOT pilot in Kuwait. The intervention employed addressable switches for selective perforation and a Single-Phase Retarded Inorganic Acid System (SPRIAS), chosen for its superior wormholing

performance and minimal precipitation risks. This workflow reduced the expected five CT runs to three, significantly streamlining the operation.

The deployment of CT HFOT led to a 68-hour reduction in rig-based operational time, delivering notable cost savings and improved HSE outcomes by reducing heavy equipment handling and personnel exposure. Most importantly, the intervention revitalized a stagnant well, achieving a remarkable 400-fold increase in production. Post-stimulation performance confirmed sustained flow rates, enhanced reservoir connectivity, and effective formation stimulation.

This study presents a detailed overview of the intervention workflow, including job design, pre-treatment diagnostics, operational execution, and post-job analysis, with comparative metrics against conventional practices. The first successful field trial of CT HFOT in Kuwait has proven its scalability and applicability for mature field rejuvenation, offering a cost-effective, efficient, and technically robust solution for reactivating shut-in wells. The learnings and outcomes of this intervention establish a new benchmark for intervention efficiency and production recovery in North Kuwait, with broader applicability across similar reservoir environments in the region.

Introduction: Technical Overview of the Mauddud Reservoir — Raudhatain Field, North Kuwait Field History & Development Context

Discovered in the 1950s and brought onstream in 1957, Raudhatain (Fig. 1) initially drew most volumes from the Lower Burgan, while the Mauddud contributed under natural depletion because the aquifer is weak. As the field matured, attention shifted to Mauddud due to its aerial extent and stacked pay potential. Before scaling development, targeted pilots in the late 1990s gathered a full static–dynamic picture—rates, pressures, production logs, and well-to-well tracer responses—to see how fluids actually move in this rock.

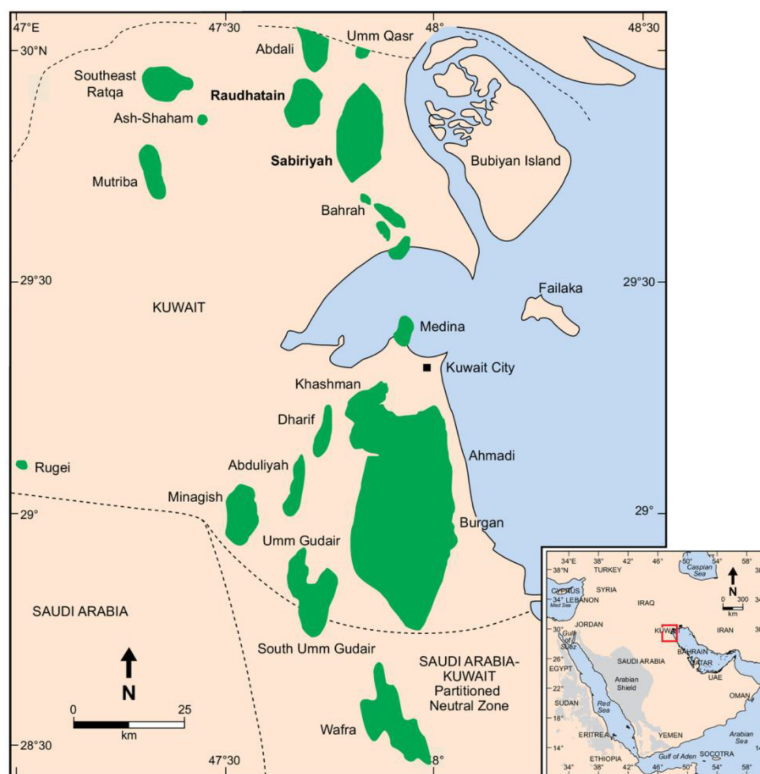


Figure 1—Arabian region showing Kuwait and the location of the Raudhatain Fields.

Two lessons have steered planning ever since: first, connectivity changes abruptly over short distances because thin, very permeable layers can shortcut flow while nearby baffled intervals slow it down; second,

performance depends less on the map label of a zone and more on where a wellbore intersects the true flow units. In practice, that history argues for designing placement and completions around heterogeneity from day one—so we stabilize inflow, avoid early water at the well scale, and keep lift systems operating in a predictable window rather than firefighting after breakthrough.

Geological & Structural Framework

Mauddud at Raudhatain (Fig. 2) is a mid-Cretaceous, shallow-marine ramp carbonate draped over a gentle dome and cut by strike-slip-related faults. The pay is organized into ten stacked zones (MaA–MaJ) that correlate regionally, but flow is governed at a finer scale by lithofacies associations within those zones. Gross thickness is roughly 350–400 ft. Storage is carried mainly by clean to clay-bearing packstones with intergranular pores, whereas grainstone and skeletal-packstone streaks act as fast channels when intersected.

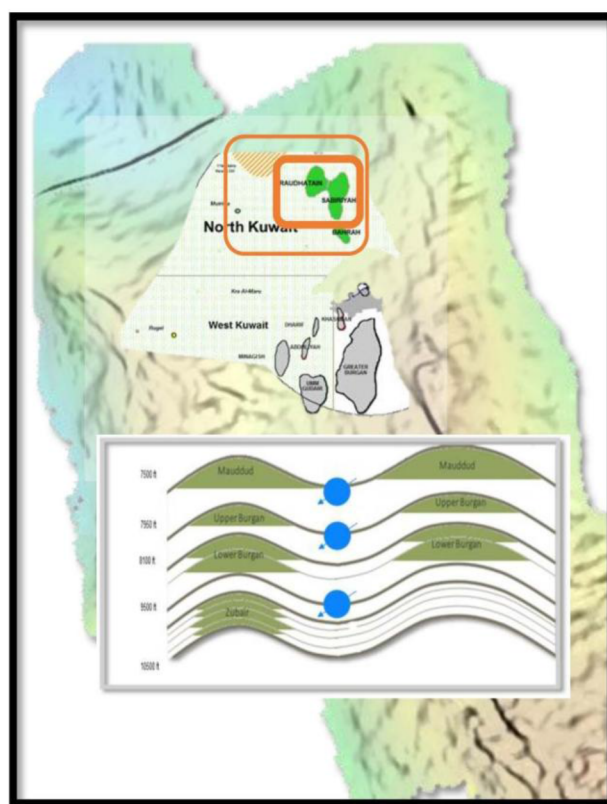


Figure 2—Raudhatain fields location in the Kuwait map

Two thin, laterally persistent features are repeatedly impactful: a rudist-rich floatstone near the upper part of the reservoir (around MaC) and a shoreface-sand streak in MaH; both can behave as thief conduits if not managed. Counterbalancing them are baffles and barriers—bioturbated wacke/packstones above MaG and within MaD/E/F, plus a field-wide mud-rich layer above MaI—that compartmentalize vertical flow and create patchy communication between layers. Structurally, faults and occasional fracture corridors locally enhance connectivity along strike while sealing across it in places. The practical takeaway is simple: this is a laterally extensive reservoir with sharp local contrasts, excellent capacity, and little tolerance for trajectories or completions that ignore thin high-perm beds and internal baffles.

Petrophysical & Fluid Characterization in Mauddud Formation

Porosity is typically 16–22%, predominantly intergranular, with only spotty secondary pores. Permeability is highly scattered—from about 0.01 mD up to roughly 1.5 D—while the dominant reservoir facies commonly fall around 30–70 mD, punctuated by localized streaks that are much higher. Initial oil saturation is high (≈ 70 –90%). Fluids follow structure: oil is lighter near the crest ($\sim 30^\circ$ API) and becomes heavier toward the flanks (mid-teens API), so mobility degrades down-structure. The reservoir was discovered moderately undersaturated (initial pressure $\sim 3,700$ psi at $\sim 7,300$ ft ss; bubble point $\sim 1,750$ psi). Vertical permeability and capillary layering are uneven, so crossflow between units is not guaranteed; thin beds can dominate production until drawdown is balanced. Importantly, Mauddud (Fig. 3 and Fig. 4) is a mid-Cretaceous (Late Albian) transgressive carbonate ramp deposited after the Burgan/Nahr Umr clastic episode and later overlain by the Wara clastic pulse. Internally, it is built from stacked transgressive–regressive cycles separated by flooding surfaces, which generate laterally persistent high-permeability sheets interbedded with baffling layers. This Cretaceous stratigraphic architecture explains the extreme k scatter, the propensity for thief zones to short-circuit displacement, and the engineering need for segmented horizontals with ICDs to equalize inflow and stabilize well performance.

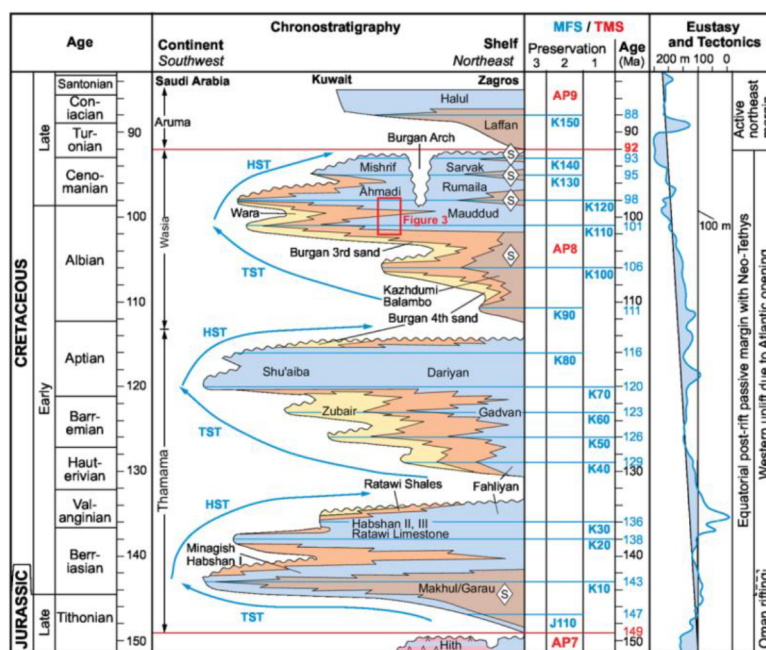


Figure 3—Mid Cretaceous Chronostratigraphy - Mauddud Formation

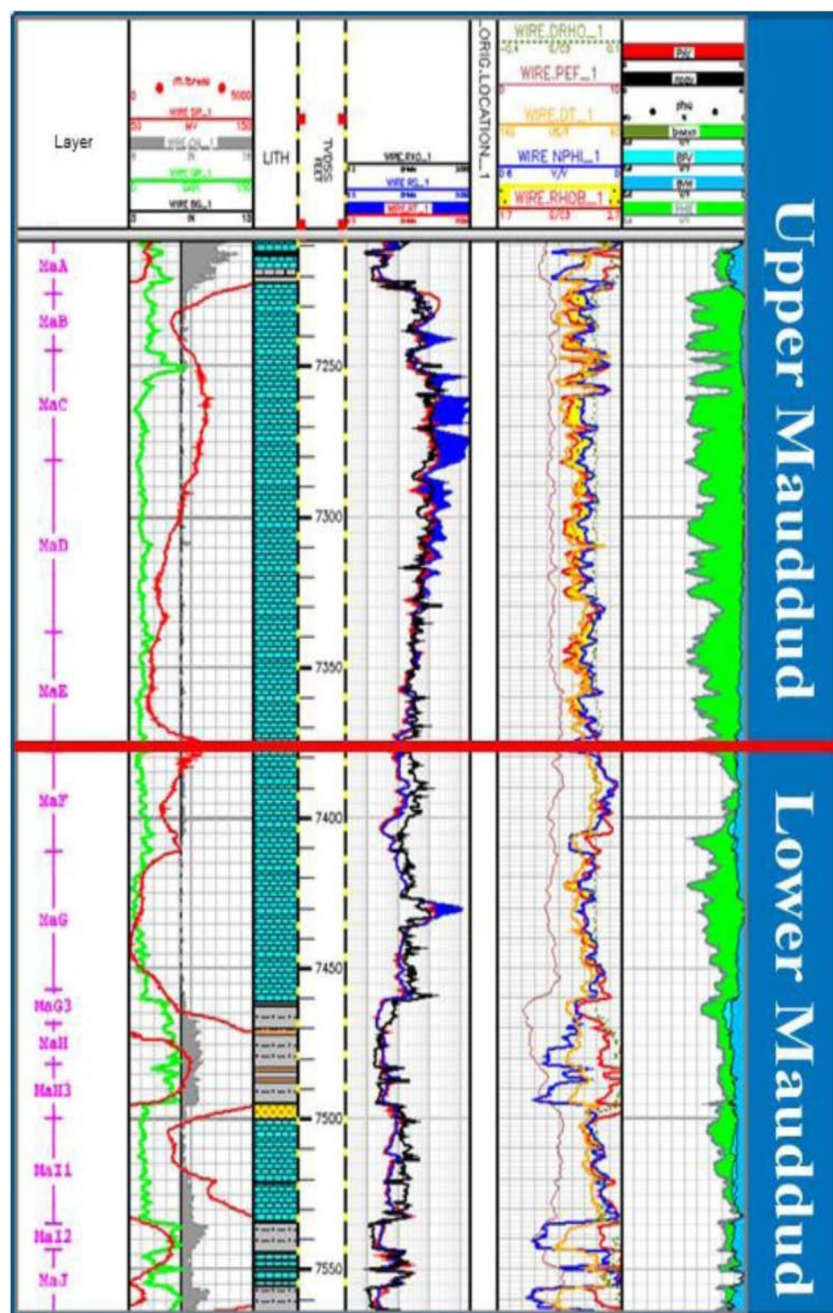


Figure 4—Raudhatain Mauddud reservoir layers

Production Concept & Planning — Horizontal Wells with ICD Completions (PHIL-Oriented)

Given the rock fabric above, the most reliable way to produce Mauddud at stable rates with controlled water is to make horizontal wells with inflow-control devices (ICDs) and swell packers the baseline completion. The placement logic is straightforward: land producers higher in the section (for example, in MaB/MaC) to harvest the oil column while avoiding the most active thief conduits; where pressure support is required, place injectors deeper (around MaE) and mechanically isolate any upper intervals known to feed high-perm streaks. Segmenting the lateral with packers and sizing ICD nozzles to introduce deliberate pressure drop across each compartment flattens inflow, counters heel dominance, and prevents one "too-good" bed from overwhelming the well. For legacy verticals, sidetracking to horizontal laterals generally delivers more

predictable inflow than adding perforations, which often accelerates water. This plan is easy to operate and easy to surveil: routine PLTs to confirm which compartments contribute, simple pressure-transient tests to check communication and skin, and targeted tracers when connectivity is uncertain. In short, design the well around the rock—produce high, inject low where needed, and use segmented horizontals with ICDs to turn a heterogeneity-limited carbonate into a predictable, lower-water-cut development with steadier ESP intake and more forecastable field behavior.

Hybrid Fiber Optics Technology (HFOT) for Coiled Tubing — Architecture, Power, and Field Execution - Scope and Rationale

Hybrid Fiber Optics Technology (HFOT) integrates an electro-optical cable within the coiled-tubing (CT) string to deliver wireline-class, real-time telemetry and continuous surface power during intervention. In practical terms, this removes battery changeouts, collapses dual-unit logistics, and allows long, multi-mode programs—production logging, stimulation, conformance, diagnostics—to run as a single continuous operation with higher data density and tighter control. The approach preserves standard CT hydraulics and reach while elevating the conveyed toolstring to a powered, instrumented system capable of live decision-making.

The backbone of HFOT is a 1/8-in OD electro-optical cable (Fig.5) qualified for high-pressure/high-temperature environments (approximately 17,500 psi and 350 °F) and deployable in common CT sizes from 1½ to 2⅞ in. The construction couples optical fibers for high-bandwidth downhole telemetry—and, when required, distributed temperature/acoustic sensing (DTS/DAS)—with insulated copper conductors sized to deliver regulated DC power from surface. A metallic serve/shield provides crush, abrasion, and fatigue resistance under repeated CT cycling and pumped operations. Engineering sizing is done to tool-head voltage at temperature, accounting for conductor resistance over the full string, thermal derating, and peak current; optical design preserves a ≥ 3 dB two-way margin after splices and connectors. Minimum bend radii at the guide arch and levelwind are respected, and termination hardware is pressure-qualified above the operating envelope to avoid cable flooding during upset.



Figure 5—Hybrid Fiber Optics Technology cable setup

HFOT's surface stack comprises three modules that operate as a single system. The Pressure Bulkhead (PBH) provides the fluid-to-air interface between the pressurized CT bore and surface equipment and terminates the hybrid cable by routing high-voltage DC toward the power unit and optical fibers toward the telemetry interface or an optional fiber breakout for DTS/DAS. The Surface Power Module (SPM) supplies continuous, adjustable, galvanically isolated DC to the downhole toolstring (Fig. 6) with soft-start, current limiting, insulation-resistance monitoring, and live voltage/current trending. The Surface Interface Module (SIM) converts the optical downhole telemetry to IP/Ethernet for the acquisition system and handles downlink control; when distributed sensing is planned, the same optical pathway is bridged to the appropriate acquisition. Because power and communications are native to the cable, wireline-class

production logging tools and real-time intervention tools can be run on CT without remaking the head; acquisition software changes are limited to power routing and control.

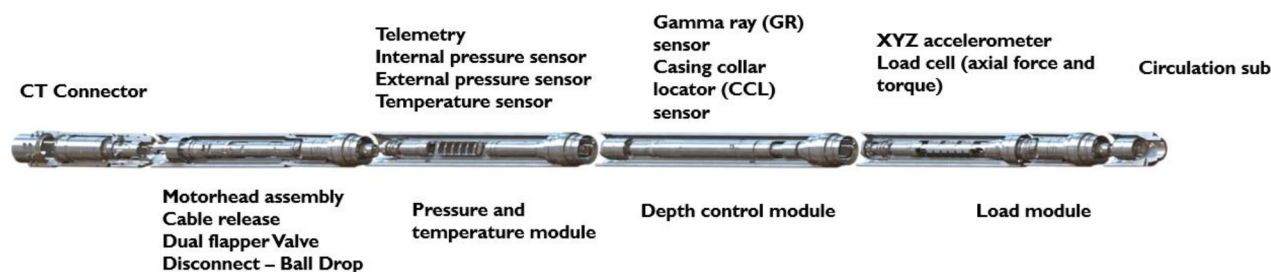


Figure 6—Hybrid Fiber Optics Technology bottomhole assembly

Downhole tools are configured primarily for surface-power operation but retain battery compatibility for contingency use. Primary telemetry runs over fiber for bandwidth and latency advantages; where toolsets support it, the copper pair can be used as a fallback communications channel. A pressure-activated release in the bottom-hole assembly allows the operator to leave the tools in hole and recover the cable and surface hardware intact via a controlled ball-drop/pressure-up sequence. The PBH maintains full-bore pumping capability, so stimulation, cleanout, and placement operations retain the standard CT operating window.

Job execution using the HFOT follows a repeatable sequence. Pre-job, electrical continuity and insulation are verified on the conductors; optical loss and reflections are checked with OTDR; the PBH is pressure-tested; and a closed-loop power/telemetry test is run between SPM/SIM and the toolstring. Rig-up mirrors standard CT with the addition of PBH and surface modules. Power-up starts with a low-voltage handshake to confirm tool boot and communications, then steps to operating voltage/current. During run-in and pumping, the operator monitors surface head voltage, current, insulation-resistance trend, and optical link quality against pre-set thresholds. If risk escalates (e.g., rising drag, stick, or unstable torque), the controlled disconnect is executed and the cable is recovered dry; fishing proceeds as a separate operation. Post-job, electrical/optical baselines are repeated and power/telemetry logs are archived to the well file.

Continuous surface power enables tool uptimes well beyond 72 hours without trips for battery service, so complex, staged workflows can be completed in one pass. Fiber telemetry delivers wireline-grade data quality, which materially improves in-situ interpretation and supports on-the-fly program adjustments (diverter schedules, valve commands, perforating logic, selective placement). Removing battery barrels shortens the BHA, reduces lifts and exposure under suspended loads, and simplifies rig-up. A single CT unit can cover both intervention and production logging, eliminating lithium battery handling and associated waste. Because tools retain battery compatibility, the same fleet can be planned for surface-powered or legacy battery operations depending on cable availability and program design.

Operating Workflow

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Performance and Planning Impact

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When engineered and verified to these criteria, HFOT converts CT from a passive conveyance into a powered, instrumented intervention platform that preserves pumping flexibility while adding wireline-grade telemetry and uninterrupted surface power. The net effect is fewer runs, lower lifting exposure, and materially better decisions made live on the well, enabling complex interventions to be executed as predictable, single-pass operations without re-architecting standard CT workflows.

Expanded Field Case Studies: Data & Operational Challenges – Field execution from North Kuwait Well RA-XY (Fig. 7 and Fig. 8) is a horizontal well drilled in the North Kuwait field and completed in 2014 with a 4 ½-inch ICD completion in a 6 1/8-inch openhole, designed to compartmentalize production from the Mauddud formation. However, due to the poor quality of the reservoir rock and the resulting low influx, the well was never brought into production and was subsequently suspended as non-productive.

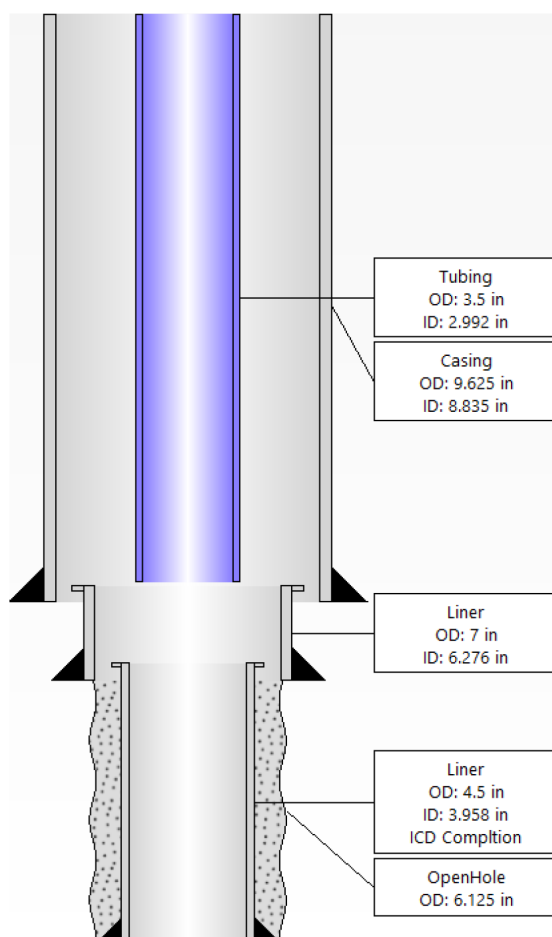


Figure 7—Well RA-XY Schematic

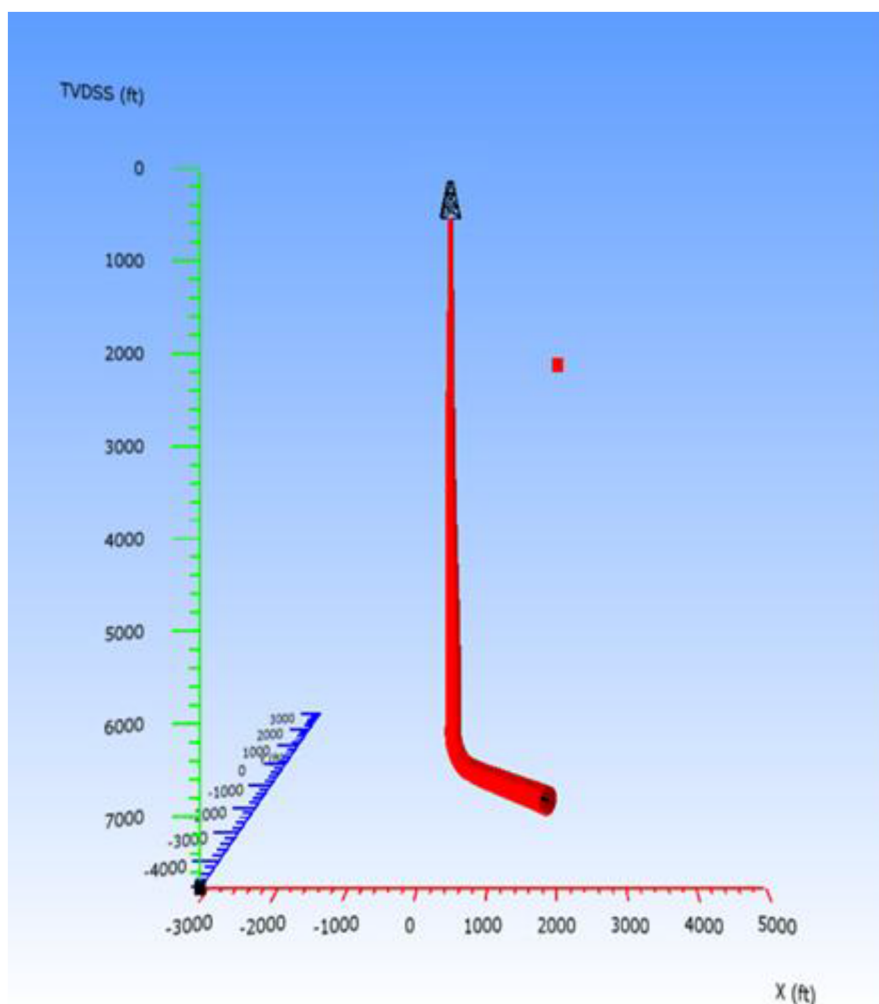


Figure 8—Wellbore directional survey in Mauddud reservoir

Consequently, the well was suspended and rendered non-productive, employing a kill string to ensure its safety. Another injector well was subsequently drilled in close proximity to Well RA-XY to provide pressure support. Since its commissioning in 2019, the new injector has been injecting water into the reservoir.

To monitor the impact of the newly drilled and commissioned injector well, periodic Shut-in Bottom Hole Pressure (SBHP) surveys have been conducted. The SBHP survey performed in 2019 indicated a pressure of ~3,500 psi. Another SBHP survey was conducted in 2021, indicating an increase in reservoir pressure to ~4,500 psi. This clearly indicates communication between the Well RA-XY producer and the new injector. Notably, recent observations have reported a wellhead shut-in pressure of ~2,000 psi at the surface. In response to controlling the pressure, the new injector was closed off, while closely monitoring the pressure on Well RA-XY. The shut-in surface pressure gradually decreased to 1,600 psi.

In 2022, the trapped pressure was released to the pit to examine any potential flows. However, no flow was observed at the surface despite opening the well at full choke and pressure dropping to 0 psi. This serves as an indication of poor rock quality, resulting in an inability to sustain flow.

In 2023, another attempt was conducted to release the buildup pressure and test the well. Despite fully opening the choke, no flow was observed at the surface, and the pressure declined to zero psi. This confirmed the presence of poor reservoir quality, with the formation unable to sustain production.

To enhance reservoir contact, a decision was made to perforate the ICDs, followed by an extensive stimulation and retesting of the well using an ESP.

A rig was mobilized to enable perforation and stimulation operations through coiled tubing. The planned intervention will be executed in five runs consisting of a drift run to proof and condition the well for the perforation runs as in (Fig.9), successfully the CT managed to reach the well end without any obstructions.

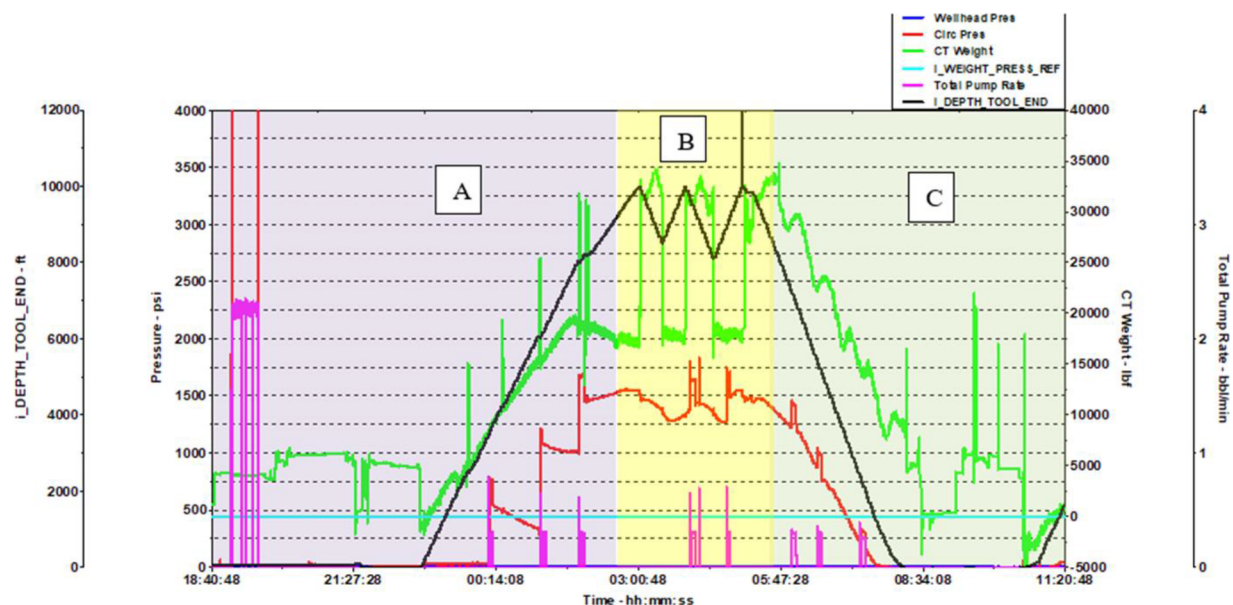


Figure 9—Run#1 to drift the well to TD

Three perforation runs were strategically designed and executed (Fig. 10) using the ASFS to achieve accurate perforation across the challenging Maudud formation to maximize the production. The plan targeted a total of 50 feet over five separate intervals within a 1,000-foot section of the well. Deep penetration guns were selected to ensure that perforation tunnels reach the formation.

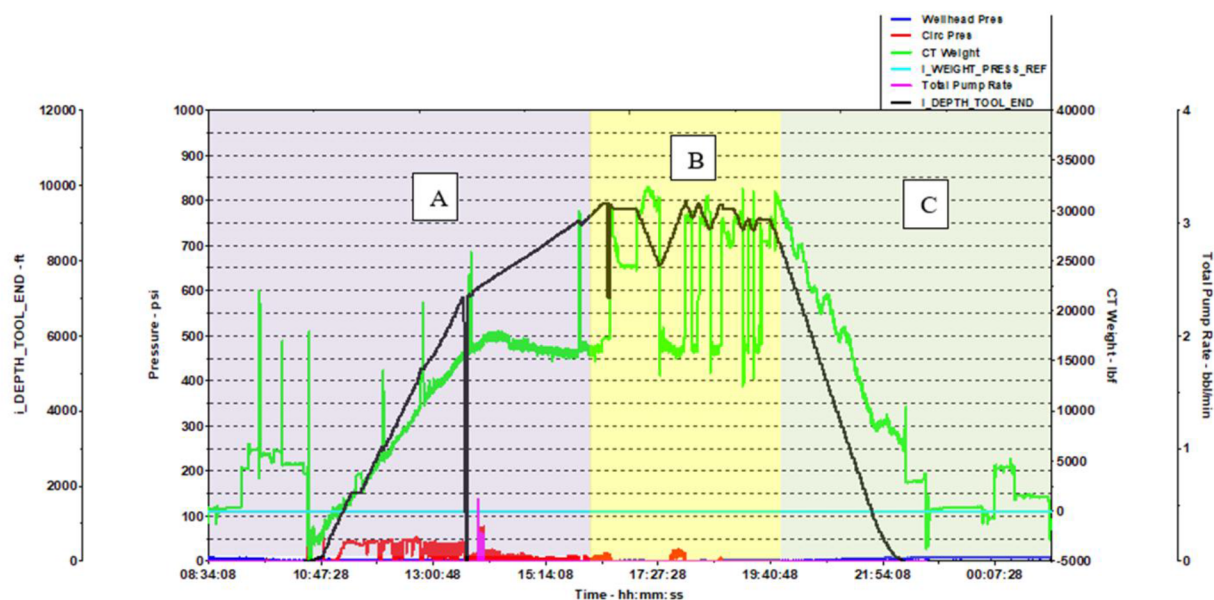


Figure 10—Run#2-4 to perform Perforation

By penetrating through the ICDs and into the reservoir, these perforations established effective flow paths for stimulation fluids, a critical step toward enhancing well productivity. Depth correlation was performed in (section B) every run as shown in Fig. 11 to accurately place the perforating guns at the correct depth

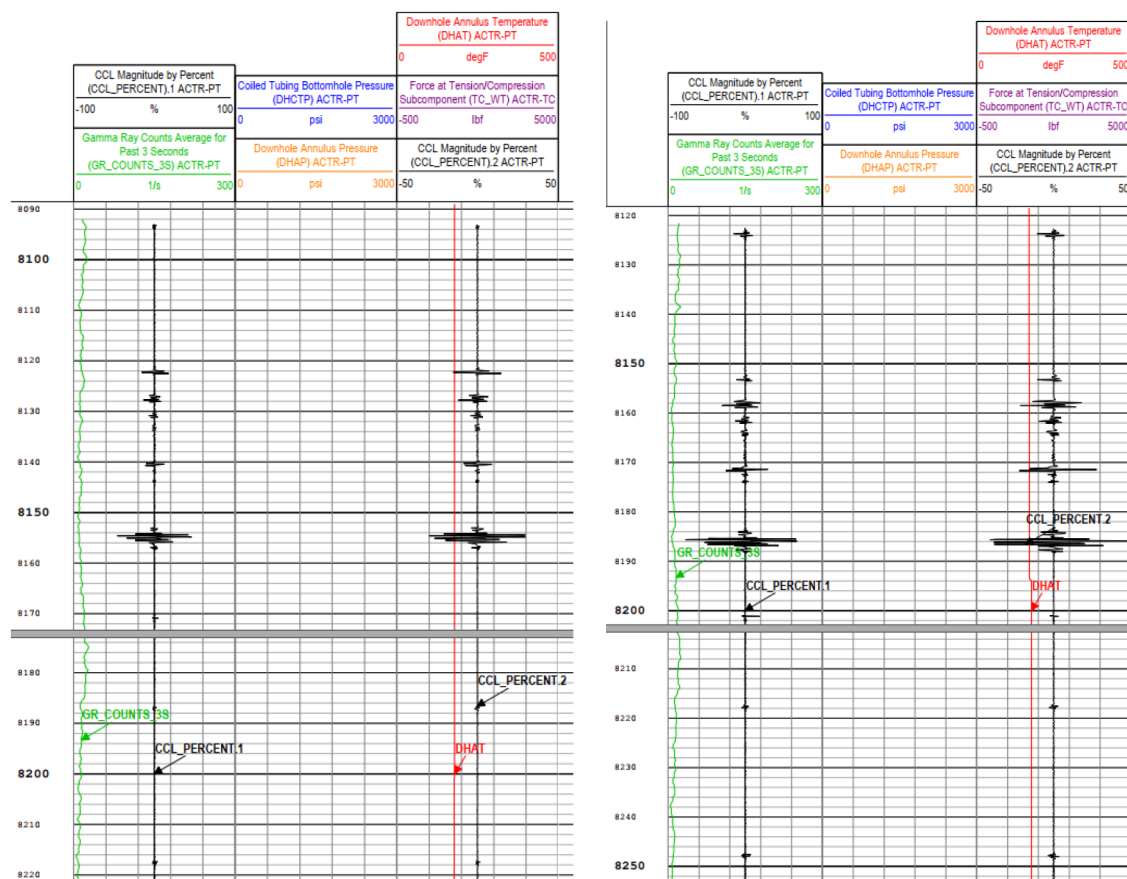


Figure 11—Depth Correlation with 30-ft offset

In the last run, Real downhole telemetry tools were deployed on CT to maintain full control during stimulation. Once CT reached the lowest perforation, the well was circulated with diesel, as shown in Section A (Fig. 12). After confirming diesel returns, the first batch of Single-phase Retarded Inorganic Acid (SPRIAS) was injected into the formation, with injection rate and pressures around 0.3 bpm and 6,000 psi simultaneously, using the hesitation method to ensure proper placement. CT was then pulled back to the end of tubing, allowing the acid to soak in the formation for three hours. Subsequently, SPRIAS and diverter treatments were pumped with CT reciprocated across the perforated interval. As the SPRIAS reacted with the formation, the pumping rate was gradually increased to maximum 2.3 bpm while maintaining injection pressures between 5,600–5,800 psi (Section B). Following treatment placement, CT was pulled out to 2,000 ft to monitor wellbore response (Section C).

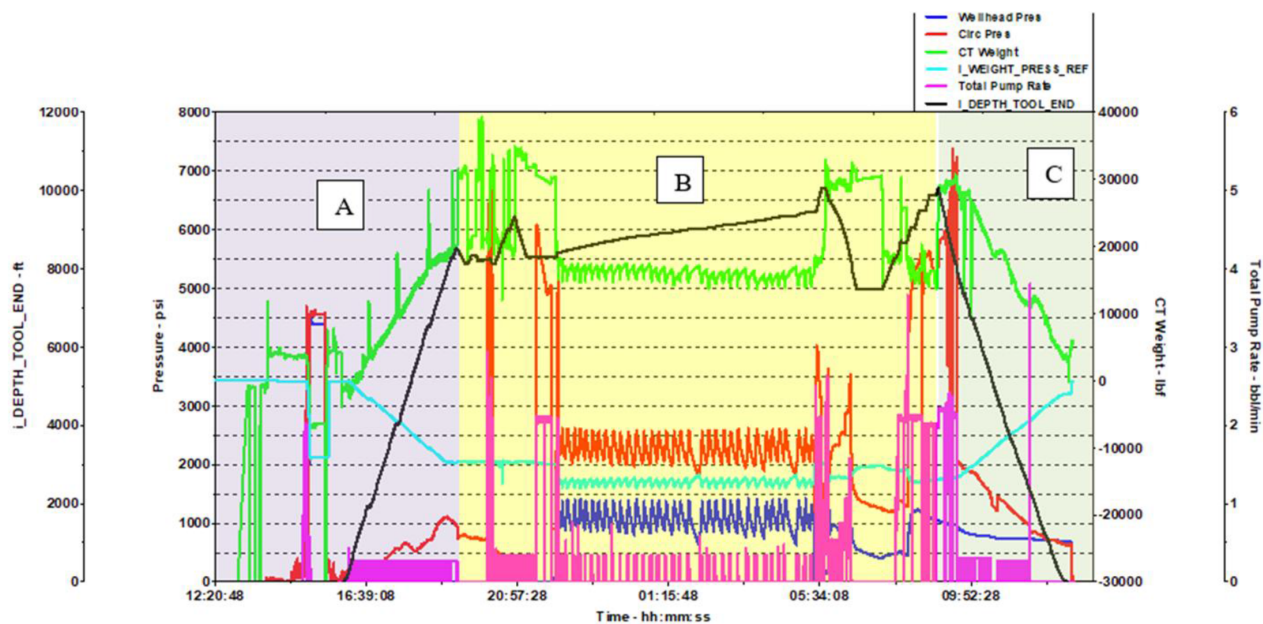


Figure 12—Run#5 to Stimulate the well

Table 1—Stimulation Schedule

Stage	Stimulation Depth (ft MD)		Net Pay	CT Pumping Rate	CT Circulation Pressure	Max Squeeze Pressure	CT Depth	Stage Fluid	Stage Volume		Cum. Volume
	From	To	ft	(bpm)	(psi)	(psi)	ft		(gal)	(bbl)	(bbl)
1	8,292	10,068	1,776	Max Rate	6,000	1,500	9,150	Pre-Flush	2,755.5	65.6	66
2	10,068	8,292	1,776	Max Rate	6,000	1,500	9,150	20% SPRIAS	1,680.0	40.0	106
POOH CT above acid level. Soak the acid for 1 hour											
3	8,292	8,692	400	Max Rate	6,000	1,500	8,092	20% SPRIAS	549.0	13.1	119
	8,692	8,692	-	Max Rate	6,000	1,500	8,692	Diverter	918.5	21.9	141
4	8,692	9,092	400	Max Rate	6,000	1,500	8,892	20% SPRIAS	549.0	13.1	154
	9,092	9,092	-	Max Rate	6,000	1,500	9,092	Diverter	918.5	21.9	175
5	9,092	9,492	400	Max Rate	6,000	1,500	8,892	20% SPRIAS	549.0	13.1	189
	9,492	9,492	-	Max Rate	6,000	1,500	9,492	Diverter	918.5	21.9	210
6	9,492	9,892	400	Max Rate	6,000	1,500	9,292	20% SPRIAS	549.0	13.1	224
	9,892	9,892	-	Max Rate	6,000	1,500	9,892	Diverter	918.5	21.9	245
7	9,892	10,068	176	Max Rate	6,000	1,500	9,980	20% SPRIAS	323.9	7.7	253
8	10,068	8,292	1,776	Max Rate	6,000	1,500	9,150	Post-Flush	2,755.5	65.6	319

Conclusions and Forward Strategy — Well RA-XY Hybrid-Enabled Rehabilitation

Well RA-XY's restart after eight years of inactivity was achieved by treating the intervention as a single, engineered system rather than a sequence of standalone tasks. HFOT on CT provided continuous surface power and real-time telemetry, allowing diagnostics, perforating, stimulation, and verification to proceed under one rig-up without breaking the data stream. The addressable-switch firing system compressed the perforating sequence into fewer, precisely controlled events, removing idle time and reducing conveyance risk. A SPRIAS overcame poor injectivity by delaying reaction and promoting deeper penetration into the near-wellbore, restoring connectivity where conventional acid would have spent prematurely. The result was a coordinated workflow in which power, data, and chemistry were aligned to a common objective: stable inflow with minimal exposure and minimal trips.

The consolidated program reduced the number of required interventions from five to three and removed approximately sixty-eight hours of rig time compared with a multi-rig-up approach. Fewer lifts, fewer pressure tests, and fewer break-outs materially lowered personnel exposure while simplifying logistics at the wellsite. During stimulation, the operation sustained a pumping rate of about 2.4 bbl/min through the CT string while maintaining downhole sensing—an internal performance best for this configuration. Most importantly, post-job production stabilized at roughly 400 BOPD, converting a long-dormant asset into a contributor without additional mechanical workovers. The perforation phase marked the first application of addressable switches on CT in the country, demonstrating that wireline-class control can be delivered on CT when powered and instrumented via HFOT.

Operational Lessons Learned and Strategic Path Forward

Three themes emerged that should shape future candidates. First, single-rig-up execution is not merely an HSE improvement; it is a performance lever. Keeping the string powered and instrumented end-to-end eliminated decision latency and avoided the restart penalties inherent to serial rig-ups. Second, perforating control matters as much as perforating design. The addressable-switch firing system removed timing uncertainty and enabled clean, verifiable sequences without tool changes, which cut conveyance cycles and protected the wellbore from unnecessary surge events. Third, stimulation efficiency is governed by placement discipline, not just acid volume. The retarded inorganic system improved injectivity because it matched the rock's kinetics; the real-time pressure/temperature feed confirmed effective placement before stepping rates or volumes. Collectively, these elements replaced trial-and-error with measured, closed-loop control.

The recommended path is to institutionalize this integrated template as the default for complex restarts and selective stimulations. HFOT should be specified as the baseline conveyance where real-time decisions can change outcome—production logging while pumping, addressable perforating under CT, and placement jobs requiring live downhole feedback. ASFS should be embedded as the standard perforation control in CT deployments to condense runs and de-risk timing. For acid systems, a screening matrix should be adopted that favors retarded inorganic formulations when injectivity and near-wellbore damage are the dominant limiters; laboratory kinetics and diversion modeling should be tied directly to field rate/pressure envelopes derived from the HFOT data. Program governance should shift from job-count metrics to outcome metrics—hours saved per well, exposure hours reduced, percentage of single-pass completions, and post-job stability of ESP intake pressure and water cut. On the readiness side, pre-job acceptance should include electrical (conductor resistance, insulation resistance), optical (OTDR baseline, attenuation budget), and hydraulic/fatigue checks for the CT-cable system, with those baselines repeated post-job to build a reliability dataset for the fleet.

Candidate wells that benefit most share three traits: long shut-in history with intact mechanical barrier envelopes, evidence of near-wellbore impairment or bypassed interval, and geometry or debris that make e-line suboptimal yet demand wireline-grade control. Applying the same integrated workflow to this subset is expected to replicate the time savings (on the order of tens of rig hours per well), maintain the reduction in rig-up/down exposure, and increase the probability of single-pass success. At portfolio scale, codifying HFOT + ASFS + chemistry selection into a single operating standard enables consistent planning, faster learning cycles, and clearer performance forecasting, while the continuous datasets captured in each job shorten the loop between surveillance and action on subsequent wells.

Here is a table summarizing most of the key achievements and the tangible Benefits Summary (Single Rig-Up CT + Hybrid Electro-Optical Intervention Workflow)

Benefit area	Baseline / Plan	Actual / Result	Quantified delta	Technical implication
Rig-up count & HSE exposure	Multiple rig-ups (equipment spotting, pressure tests, rig-down between phases)	Single rig-up covered the entire operation	Fewer exposure cycles (notional 3–4× fewer pressure tests/rig-downs)	Lower cumulative HSE risk; fewer integrity re-tests; less NPT from reconfiguration
Rig-up duration	~24 h (initial discussion)	18 h	–6 h (–25%)	Faster time-to-operations; earlier RIH
Slack-management setup	Planned 24 h (to accommodate rate limits on CT)	12 h	–12 h (–50%)	Constraint cleared sooner; greater schedule certainty
Perforating run duration	15–16 h per run (typical)	11 h per run (RIH, correlate, shoot, POOH)	–4 to –5 h per run (≈–25 to –31%)	Higher cycle efficiency with real-time control/telemetry
Number of perforating runs	5 runs (initial plan)	3 runs	–2 runs	Fewer conveyance cycles and pressure tests; lower risk/exposure
Perforating time vs. plan (aggregate)	5 × 15–16 h = 75–80 h	3 × 11 h = 33 h	–42 to –47 h saved	Material rig-time reduction purely from run count + faster cycles
Injectivity during matrix treatment	Poor injectivity; WHP 1,500 psi @ 0.3 bpm ; BHP ~5,850 psi	Completed at 2.4 bpm with 990 psi WHP ; BHP ~5,580 psi	8× rate increase with ~510 psi WHP reduction; ~270 psi BHP drop	Clear injectivity gain while monitoring BHP in real time; safer squeeze window
Surface/BHP surveillance	Intermittent or inferred	Real-time WHP/BHP trending throughout treatment	Continuous	Confident rate-stepping; immediate response to formation behavior
Overall injectivity outcome	—	“Overall enhanced injectivity” (post-treatment)	Qualitative gain	Supports sustained higher placement rates post-job
Production outcome	—	~300 bbl of oil produced (post-intervention)	Tangible recovery	Confirms effective restoration of flow capacity

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We acknowledge the Well Surveillance, Rig & Workover, and Field Development teams in the North Kuwait border fields for planning, execution, and surveillance. Their field discipline—data acquisition, pressure testing, QA/QC, and operational readiness—underpinned the results presented here.

We also recognize the Reservoir Performance Intervention technical team from our service partner in Kuwait for system engineering, tool integration, and on-site support, including hybrid cable terminations, surface power/telemetry interface checks, and post-job data validation.

Finally, we appreciate the contributions of colleagues who reviewed the procedures and manuscript. Any interpretations or opinions expressed are those of the authors and do not necessarily reflect the views of the affiliated organizations.